

Yiding Luo

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Research Interests: High-impact Weather, Machine Learning, Mesoscale Meteorology

EDUCATION

University of Washington - Seattle

09/2023 - Present

- Bachelor of Science in Atmospheric & Climate Science - Meteorology
- Bachelor of Science in Atmospheric & Climate Science - Data Science
- Minor in Applied Mathematics
- Current Overall GPA: 3.81/4.0
- Expected graduation: **June, 2027**

RESEARCH & INTERN EXPERIENCES

Lead Undergraduate Researcher, **University of Washington - Seattle**

10/2024 - Present

Evaluation of Machine-Learning Weather Models in Reproducing Mesoscale Tornado-Favorable Convective Environments

- Supervisor: Prof. Alexandra Anderson-Frey
- Led a research team of two undergraduate students through the full research lifecycle, including topic selection, experimental design, model benchmarking, and manuscript writing.
- Proposed a composite verification framework (intensity, spatial, and object-based) to evaluate Transformer-based ML weather models (Pangu-Weather, FengWu, FuXi) in reproducing tornado-favorable convective environments across 24-168 h lead times.
- Revealed that extreme-value over-smoothing in ML weather models is architecture-dependent, and diagnosed model-specific internal biases through vertical-profile analysis, linking forecast errors to underlying physical mechanisms.
- Publication: **First-author** manuscript drafted; submission targeted for October 2026. Preprint forthcoming upon submission.

GIS & Meteorological Data Intern, **Sichuan Meteorological Bureau, China**

07/2025 - 08/2025

- Meteorological data visualization and mini software development
- Curated and cleansed meteorological station data from CMA Big Data and Cloud Platform (Tianqing); mapped spatiotemporal patterns using ArcGIS (shapefiles).
- Automated the monthly data-reporting pipeline from the provincial meteorological bureau to the provincial energy bureau, reducing manual processing time of a single task from 40 minutes to 15 seconds.

HONORS

2025 Sally Schoenberg and Randy Poteet Scholarship

2025-2026 academic year

2026 Sally Schoenberg and Randy Poteet Scholarship

2026-2027 academic year

2026 Mindlin Scholarship

2026-2027 academic year

SYMPOSIA & PROJECTS

ENSO-Columbia River Streamflow Analysis, **University of Washington - Seattle** 06/2026

How does the Oceanic Niño Index (ONI) influence Columbia River streamflow, and what are the mechanisms (snowpack-mediated pathway) connecting ENSO to hydrological response in the Pacific Northwest?

- Computer Science and Engineering group project (CSE 163: Intermediate Data Programming)
- Investigated ENSO-Columbia River streamflow teleconnections, building a Python pipeline for lag correlation and ENSO composite analysis to quantify the snowpack-mediated hydrological pathway.

Research Symposia (presenter - first author), **University of Washington - Seattle** 05/2025 & 05/2026

ML-Based Weather Models' Performance Evaluation

- Departmental Research Symposium (May 2025) — Oral/Poster Presentation in the department
- The 28th Annual Undergraduate Research Symposium (May 2026) — Campus-wide Poster Presentation. Conceptualized the study design, conducted all data analysis, and generated all figures and visual content for both presentations. Improved and updated from last year's results and poster.
- Departmental Research Symposium (May 2026) — Presented the updated poster in the department

Paper Reproduction: ENSO Recharge Oscillator, **University of Washington - Seattle** 06/2025

Reproduced *LIU et al. (2018, Appl. Math. Mech., doi: 10.21656/1000-0887.380332)*

- Applied mathematics course project (AMATH 383: Introduction to Continuous Mathematical Modeling)
- Derived all mathematical formulations independently and generated corresponding figures for validation.
- *Paper Reproduction report: **Yiding Luo** and Rui Yuan (2025). Analyzing the bifurcation of the ENSO replenishment oscillator with delayed feedback. AMATH 383, University of Washington.*

EXTRACURRICULAR

DawgCast, **University of Washington - Seattle** 10/2024 - 06/2026

- Campus-wide weather forecast blogging club
- Proficient in MAX (Meteorological Analysis and Visualization Software) for operational weather analysis in the Northwest Pacific.

Science Communicator (Meteorology), **Bilibili Video Platform** 12/2019 - 06/2025

- Create animated educational videos explaining atmospheric science concepts for public audiences.

Skills

- **Programming:** Python (netcdf, numpy, scipy, xarray, pandas, etc.), SQL, R, LaTeX, Markdown, ArcGIS
- **Meteorology:** mesoscale meteorology, synoptic analysis, ingradient-based forecasting, convective environment Forecasting, ensemble forecasting, ENSO
- **Machine Learning:** PyTorch, basics of Transformer, ViT, CNN, LSTM, diffusion model, LORA, PINN
- **Statistics:** RMSE, ACC, t-tests, logistic regression, Chi-Square, performance diagram, forecast verification
- **Tools & Platforms:** NCAR Derecho supercomputer, CUDA, VS Code, PyCharm, RStudio, HPC job scheduling, MAX, Adobe Premiere Pro, Adobe Photoshop